

P8131 Homework 9

Yongyan Liu (yl6107)

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Problem 1

Determine the survival and density functions for a continuous survival time variable with hazard function

$$h(x) = \frac{2x}{1+x^2}, \quad x \geq 0.$$

Hint: consider the derivative of $\log(1+x^2)$.

From Lecture 21 the survival function is recovered from the hazard via

$$S(x) = \exp\{-H(x)\}, \quad H(x) = \int_0^x h(u) du,$$

Using the hint, observe that h is exactly the derivative of $\log(1+u^2)$.

$$H(x) = \int_0^x \frac{2u}{1+u^2} du = \left[\log(1+u^2) \right]_0^x = \log(1+x^2).$$

$$S(x) = \exp\{-H(x)\} = \exp\{-\log(1+x^2)\} = \frac{1}{1+x^2}, \quad x \geq 0.$$

Using $f(x) = h(x)S(x)$,

$$f(x) = \frac{2x}{1+x^2} \cdot \frac{1}{1+x^2} = \frac{2x}{(1+x^2)^2}, \quad x \geq 0.$$

Problem 2

For the data 1, 2, 2, 4⁺, 5⁺, 6, 7⁺, 8⁺, 9⁺, 10⁺ (where ⁺ denotes right censoring), write out the data table and calculate (a) the Kaplan–Meier estimate of $S(t)$, (b) the Nelson–Aalen estimate of $H(t)$, and (c) the Fleming–Harrington estimate of $S(t)$.

See the table below:

t_i	d_i	c_i	n_i	$\hat{\lambda}_i = d_i/n_i$	$\hat{S}_{\text{KM}}(t_i)$	$\widetilde{H}(t_i)$	$\widehat{S}_{\text{FH}}(t_i)$
1	1	0	10	1/10 = 0.1000	0.9000	0.1000	0.9048
2	2	0	9	2/9 ≈ 0.2222	0.7000	0.3222	0.7245
4	0	1	7	0	0.7000	0.3222	0.7245
5	0	1	6	0	0.7000	0.3222	0.7245
6	1	0	5	1/5 = 0.2000	0.5600	0.5222	0.5932
7	0	1	4	0	0.5600	0.5222	0.5932
8	0	1	3	0	0.5600	0.5222	0.5932
9	0	1	2	0	0.5600	0.5222	0.5932
10	0	1	1	0	0.5600	0.5222	0.5932

Problem 3

Use the tongue data in the R package KMsurv. For each tumor type (aneuploid and diploid), plot the Kaplan–Meier curve of the survival function and its pointwise 95% confidence intervals (using the log transformation). What are the estimated 1-year survival rate and 95% CI?

The tongue data set records survival of 80 patients with cancer of the tongue, classified by tumor DNA profile: `type = 1` is aneuploid ($n_1 = 52$) and `type = 2` is diploid ($n_2 = 28$); `time` is in weeks and `delta` is the death indicator. One year therefore corresponds to $t = 52$ weeks.

```
data(tongue)
str(tongue)
```

```
## 'data.frame': 80 obs. of 3 variables:
## $ type : int 1 1 1 1 1 1 1 1 1 1 ...
## $ time : int 1 3 3 4 10 13 13 16 16 24 ...
## $ delta: int 1 1 1 1 1 1 1 1 1 1 ...
```

```
table(tongue$type)
```

```
##
## 1 2
## 52 28
```

```
aneu <- subset(tongue, type == 1)
dipl <- subset(tongue, type == 2)

fit_aneu <- survfit(Surv(time, delta) ~ 1, data = aneu, conf.type = "log")
fit_dipl <- survfit(Surv(time, delta) ~ 1, data = dipl, conf.type = "log")
```

KM curves with log-transform 95% CIs

```
plot(fit_aneu, conf.int = TRUE, mark.time = TRUE,
     xlab = "Time (weeks)", ylab = "Survival probability",
     main = "Aneuploid (type = 1)")
abline(v = 52, lty = 3, col = "grey50")
```

```
plot(fit_dipl, conf.int = TRUE, mark.time = TRUE,
     xlab = "Time (weeks)", ylab = "Survival probability",
     main = "Diploid (type = 2)")
abline(v = 52, lty = 3, col = "grey50")
```

1-year survival rate and 95% CI

```
summary(fit_aneu, times = 52)
```

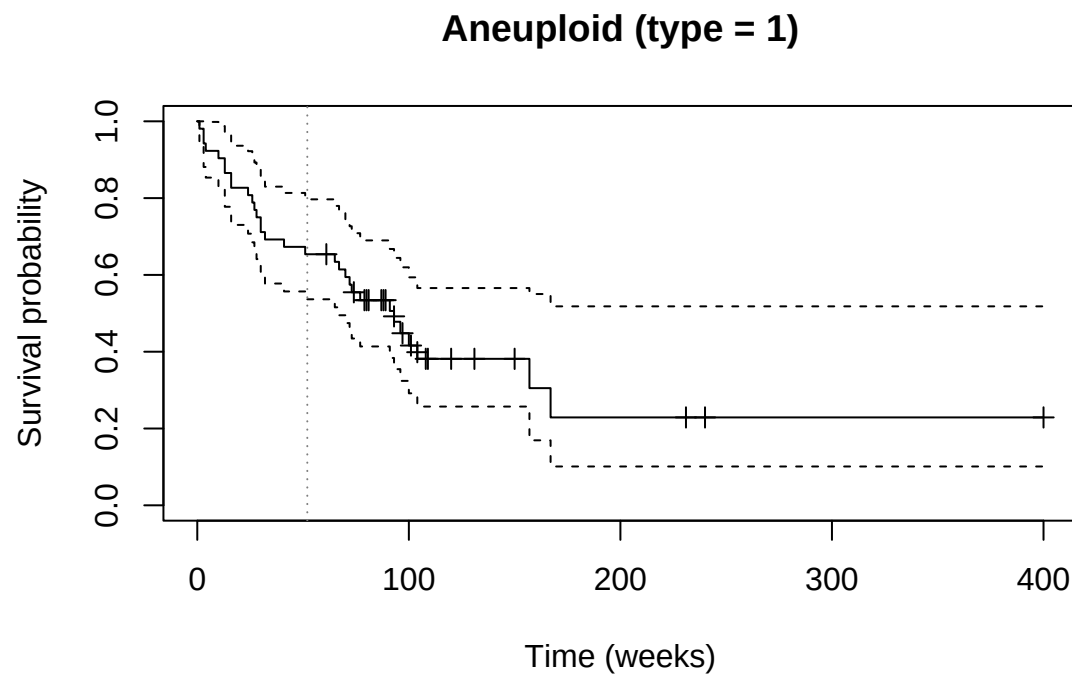


Figure 1: Kaplan–Meier survival for aneuploid tumors with pointwise 95% log-transform CIs. Tick marks indicate right-censored observations; the dotted vertical line marks $t = 52$ weeks (1 year).

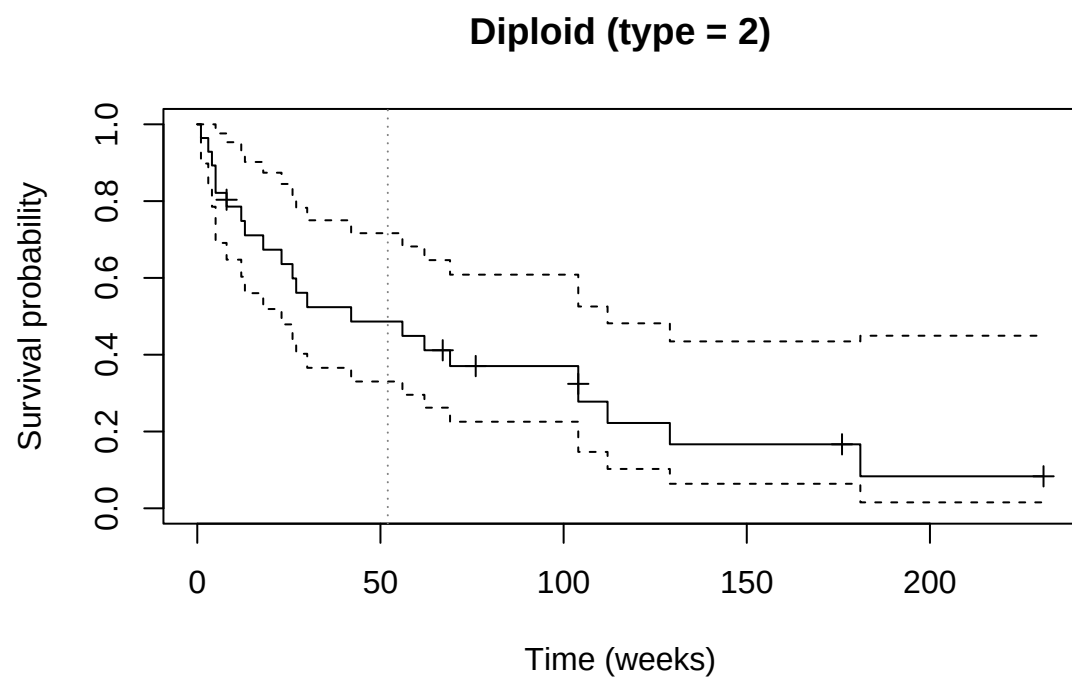


Figure 2: Kaplan–Meier survival for diploid tumors with pointwise 95% log-transform CIs. Tick marks indicate right-censored observations; the dotted vertical line marks $t = 52$ weeks (1 year).

```
## Call: survfit(formula = Surv(time, delta) ~ 1, data = aneu, conf.type = "log")
##
##   time n.risk n.event survival std.err lower 95% CI upper 95% CI
##    52    34     18   0.654   0.066    0.537    0.797
```

```
summary(fit_dipl, times = 52)
```

```
## Call: survfit(formula = Surv(time, delta) ~ 1, data = dipl, conf.type = "log")
##
##   time n.risk n.event survival std.err lower 95% CI upper 95% CI
##    52    13     14   0.486   0.0961    0.33    0.716
```

- **Aneuploid tumors (type = 1).** The estimated 1-year (52-week) survival rate is $\hat{S}(52) = 0.654$, with 95% log-transform CI (0.537, 0.797).
- **Diploid tumors (type = 2).** The estimated 1-year (52-week) survival rate is $\hat{S}(52) = 0.486$, with 95% log-transform CI (0.330, 0.716).

Patients with aneuploid tumors have a noticeably higher estimated 1-year survival probability than patients with diploid tumors, although the diploid CI is wide (smaller subgroup, $n_2 = 28$) and the two confidence bands overlap.